Seed-chain-extend alignment is accurate and runs in close to $O(m \log n)$ time for similar sequences: a rigorous average-case analysis

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Seed-chain-extend model

Random model

Extension runtime and recoverability

Chaining runtime

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- Sequence alignment – optimally solved in worst-case $O(mn)$ time
- Dynamic programming (Smith and Waterman, 1981; Needleman and Wunsch, 1970)

Problems

Consider *read alignment* – $n \approx 3,000,000,000$, $m \approx 150 - 1,000,000$

- Optimal methods are slow, so heuristics are used – BWA (Li and Durbin, 2009), minimap2 (Li, 2018), bowtie2 (Langmead and Salzberg, 2012), etc

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- Optimal methods are slow, so heuristics are used – BWA (Li and Durbin, 2009), minimap2 (Li, 2018), bowtie2 (Langmead and Salzberg, 2012), etc
- But heuristics lack theoretical guarantees

Question

Can we put non-trivial theoretical guarantees on heuristic alignment methods?

Seed-chain-extend

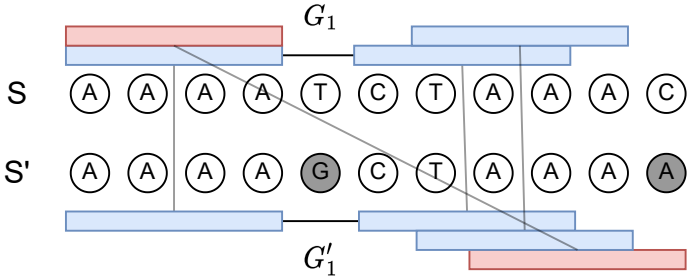
- We analyze the *seed-chain-extend* alignment heuristic

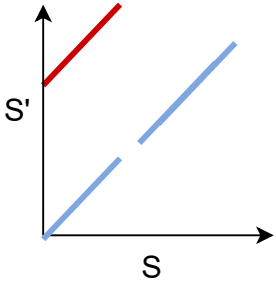
Seed-chain-extend steps

Align two strings S, S' of length n and $m \leq n$.

1. **Seeding and matching k-mers**

Seeding and matching k-mers





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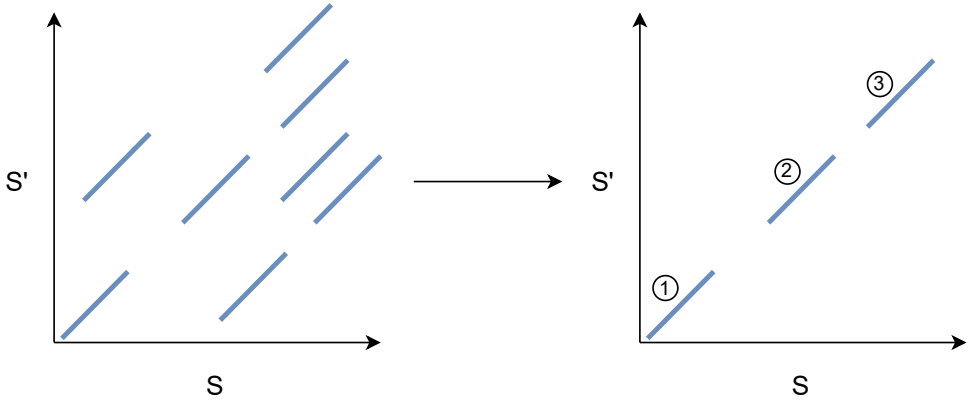

Seed-chain-extend steps

1. Seeding and matching k-mers
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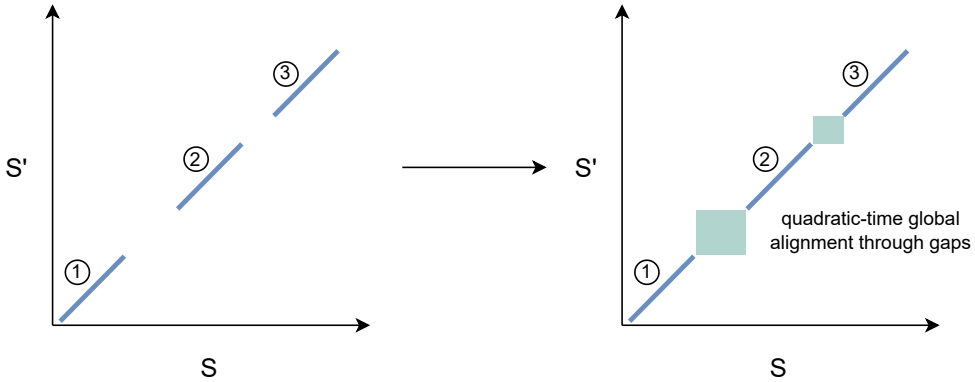
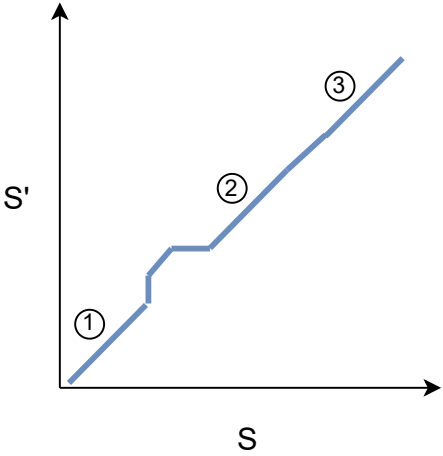
3. **Dynamic programming extension through gaps**

Dynamic programming extension through gaps



Dynamic programming extension through gaps



Seed-chain-extend use-cases

- Used in read-to-genome (e.g. minimap2) or genome-to-genome alignments (e.g. MUMmer (Marçais et al., 2018))

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- Worst-case still $O(mn)$ – how do we explain the better than $O(mn)$ behavior? (Medvedev, 2022)
- **Average-case analysis**

Random sequence model

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2. "read" S' - substring of S, length $m \leq n$, i.i.d substitutions with probability θ
3. k-mer length $k = C \log n$ for some $C(\theta) \approx 2$.

Runtime

- Runtime $= T_{Seed} + T_{Chain} + T_{Extension}$
- $T_{Seed}, T_{Chain}, T_{Extension}$ are **random variables**

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Runtime

- Runtime $= T_{Seed} + T_{Chain} + T_{Extension}$
- $T_{Seed}, T_{Chain}, T_{Extension}$ are **random variables**
- Want $\mathbb{E}[T_{Seed} + T_{Chain} + T_{Extension}]$
- Focus on $T_{Chain}, T_{Extension}$.

Extension runtime and recoverability

Two questions:

- What is $\mathbb{E}[T_{Extension}]$?
- How good is the resulting alignment?

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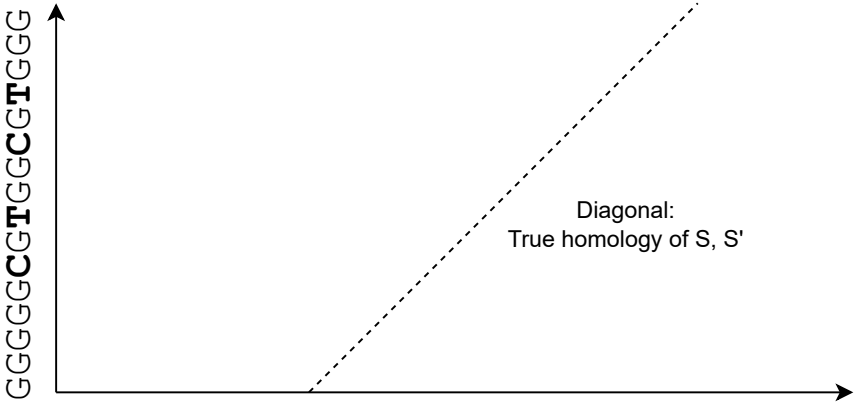
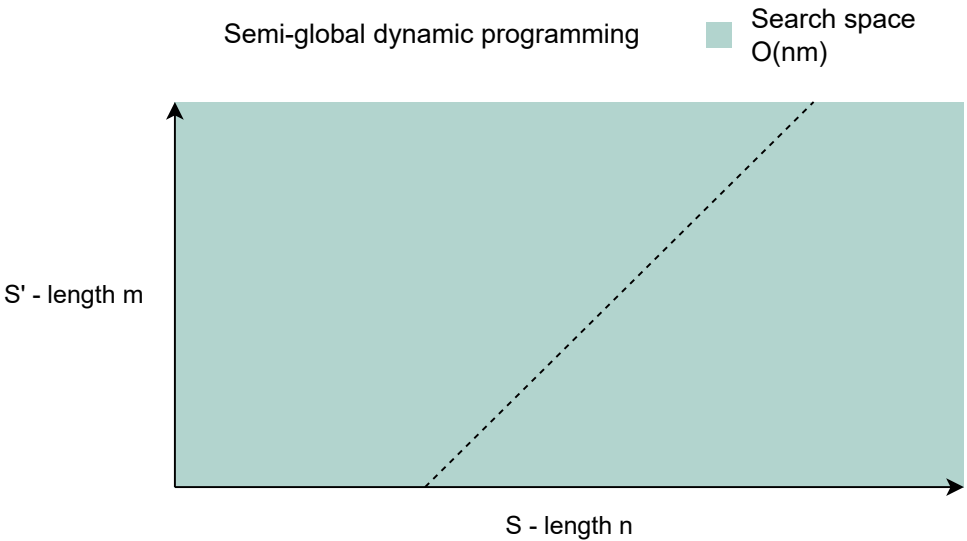
Random sequence and alignment

AAAAGGGGGGGGGGGGGGGGGGGGGGGGTTTT

GGGGG**C**G**T**GG**C**G**T**GGG



DP search space



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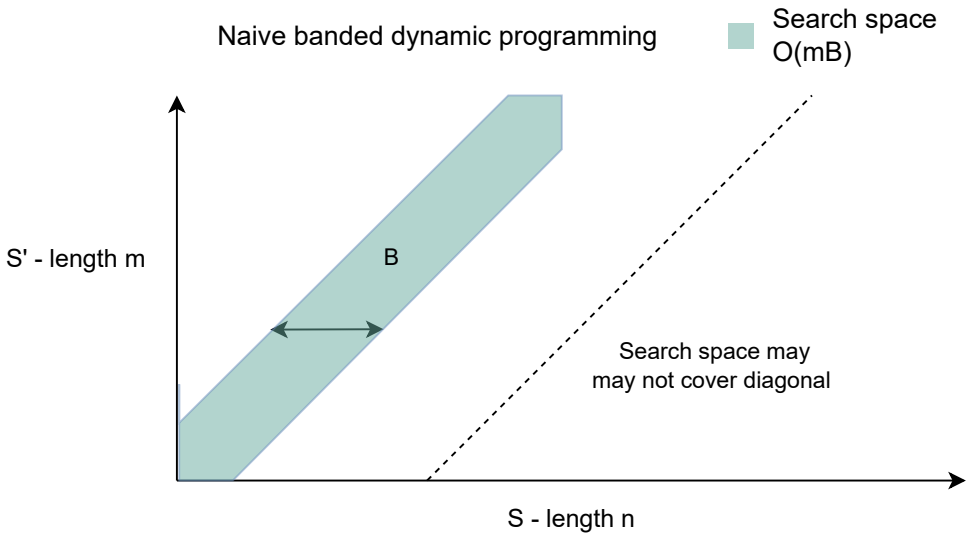
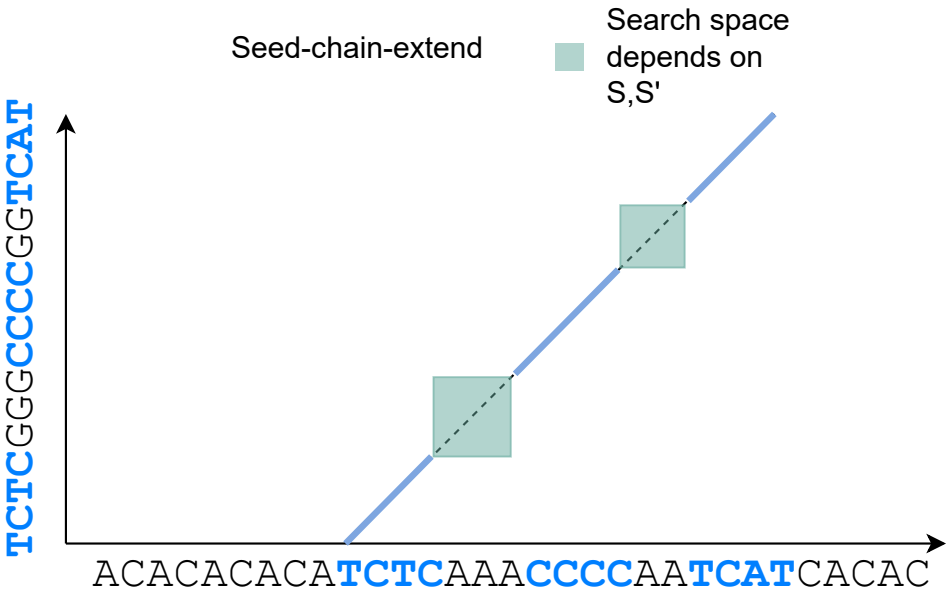
Banded DP search space



Seed-chain-extend search space



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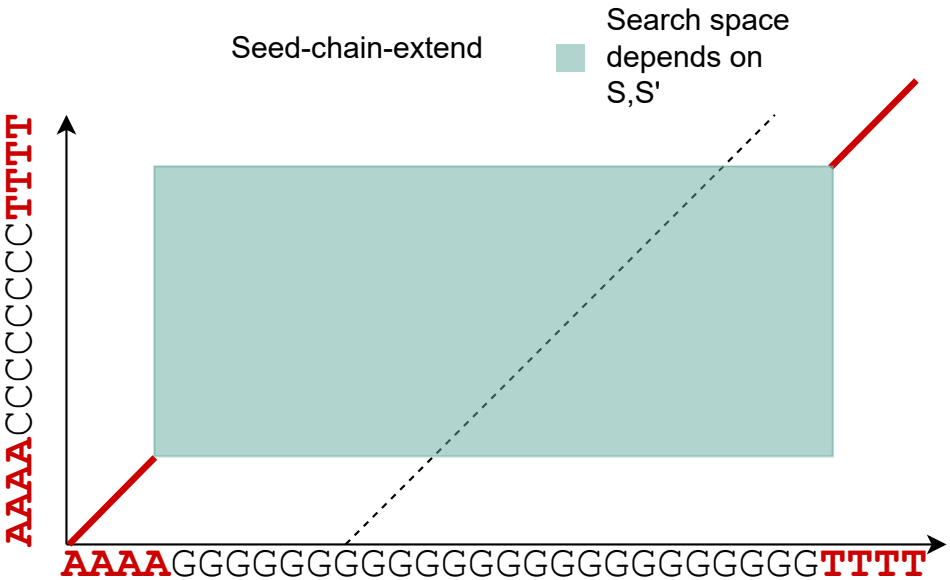

Extension runtime and recoverability

1. $\mathbb{E}[T_{Extension}]$ – **expected size search space**

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Theorem (Theorem 1 simplified from Shaw and Yu (2023))

$\mathbb{E}[T_{Extension}]$ is $O(mn^{f(\theta)} \log n)$, where $f(\theta) < 2.43 \cdot \theta$. The expected recoverability is $\geq 1 - O(\frac{1}{\sqrt{m}})$.

Proof idea

1. Sums of *almost independent* k-mer random variables $\implies$ no degenerate chainings with high probabililty (Janson, 2004; Ganesh and Sy, 2020)

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2. $\mathbb{E}[T_{Extension}] \approx \mathbb{E}[\text{gap size}^2] \leq O(m(1-\theta)^{-k} \log n) = O(mn^{f(\theta)} \log n)$

Runtime of T_{Chain}

- Let N be the number of k-mer matches or anchors (N is a random variable)

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- **Linear gap cost** objective – optimally solved in $T_{Chain} = O(N \log N)$ time (Abouelhoda and Ohlebusch, 2005; Jain et al., 2021)

Theorem (Theorem 6 simplified from Shaw and Yu (2023))

Under our random model with S length n, S' length m, $k = C \log n$,

$$\mathbb{E}[T_{Chain}] = O(m \log m)$$

Real nanopore runtimes

- Are our asymptotic results accurate?
- Took nanopore reads of $\sim 95\%$ accuracy from various species, so $\theta = 0.05$

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- Are our asymptotic results accurate?
- Took nanopore reads of $\sim 95\%$ accuracy from various species, so $\theta = 0.05$
- Aligned with custom seed-chain-extend aligner, $k \approx 2 \log n$

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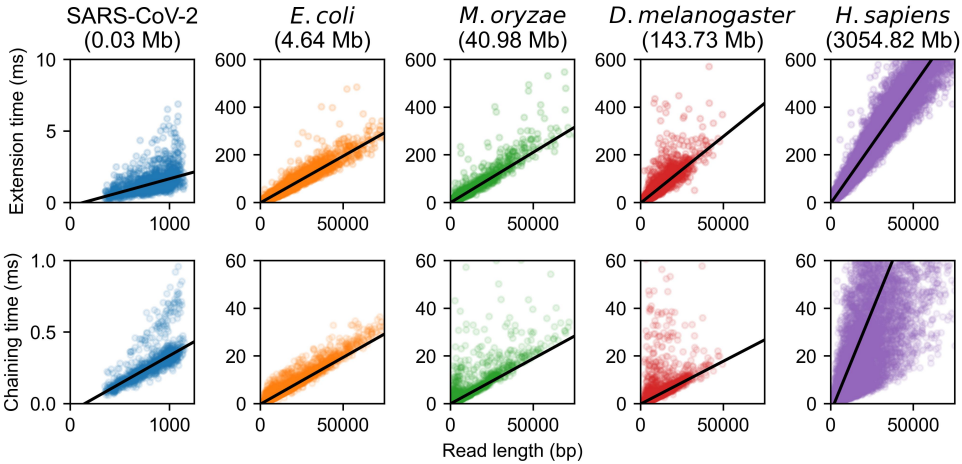
Chaining + Extension runtimes



Extension runtime

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- Slope of extension runtime $\propto n^{0.08} \log n$ (approximately)

Extension runtime predictions are informative

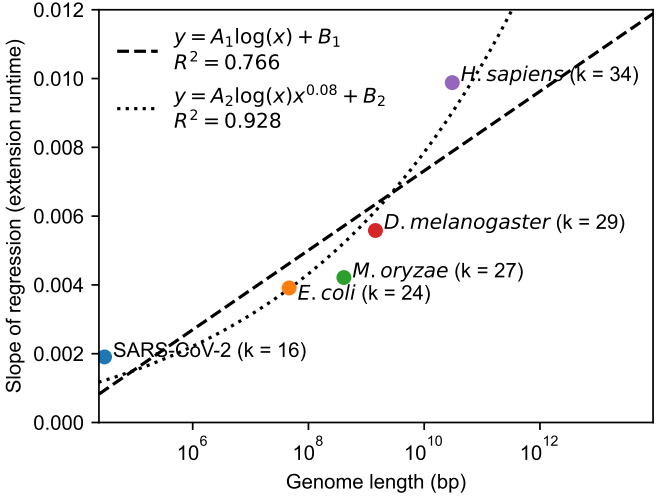

Figure: The $x^{0.08} \log x$ fitted line is better than just $\log x$ – predicted from theory!

Sketching result

- Real long-read aligners don't use all k-mers – subsample via *sketching*
- Minimizers (Roberts et al., 2004), syncmers (Edgar, 2021), etc

Sketching

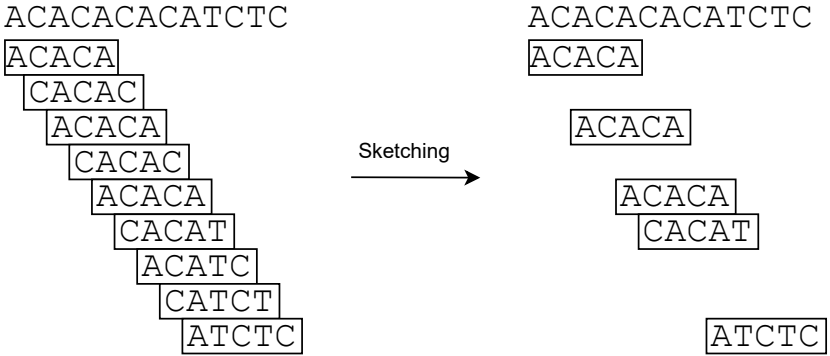

Figure: Sketching – subsampling k-mers

Sketching result

Let $c = \Theta(\log n) < k$.

Theorem (Theorem 2 from Shaw and Yu (2023) simplified)

We can subsample to $\frac{1}{c}$ of the k-mers using the open syncmer method, and

- *the* **same extension runtime/recoverability results hold**
- *but* **chaining only takes $\frac{1}{c}$ as long**

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Sketching provably **reduces** chaining time **without increasing** extension time too much!

Simulation – sketching vs no sketching

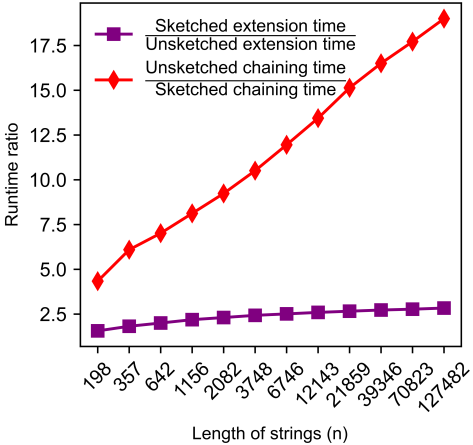

Figure: Align two length n strings with $\theta = 0.10$. Sketching with fraction $c \approx 2 \log n$

Conclusion

Motivation

- Optimal sequence alignment is slow – so we use heuristics

What we did

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- Proved $O(mn^{f(\theta)} \log n) \ll O(mn)$ runtime and good accuracy in *expectation* for seed-chain-extend.

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What we did

- Proved $O(mn^{f(\theta)} \log n) \ll O(mn)$ runtime and good accuracy in *expectation* for seed-chain-extend.
- Bounds also work for sketching and real data; provides justification usage of sketching in long-read aligners

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Proving sequence aligners can guarantee accuracy in almost O(m log n) time through an average-case analysis of the seed-chain-extend heuristic

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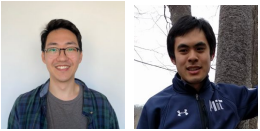

Figure: Jim Shaw, Yun William Yu

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